EVERY OBJECT IS A HOMOTOPY COLIMIT BUILT FROM A GENERATING FAMILY OF COMPACT OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11400-11429, following the statement at 11385-11398.

Editorial note. No omissions: the tower, the map to X, and the vanishing of C are all constructed explicitly; compactness enters via lemma-commutes-with-countable-sums.

Allowed standard prerequisites (editorial).

- Compact objects commute with countable direct sums. lemma-commutes-with-countable-sums (same chapter, cited around line 11417). Lets any map from E_i[m] into C factor through some X_n.
- Generation by a family. same chapter definition; the hypothesis that the direct sum of the E_i generates D. Converts the vanishing of Hom(E_i[m], C) for all i, m into C = 0.

Source reference. Original lemma 094A: sources/stacks-derived/pinned-original-source.tex, source lines 10623–10623. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category with direct sums. Let E_i, $i \in I$ be a family of compact objects of $\mathcal{D}$ such that $\bigoplus E_i$ generates $\mathcal{D}$. Then every object X of $\mathcal{D}$ can be written as*

$$X = hocolim X_n$$

where X_1 is a direct sum of shifts of the E_i and each transition morphism fits into a distinguished triangle $Y_n \to X_n \to X_{n+1} \to Y_n[1]$ where Y_n is a direct sum of shifts of the E_i.

Proof. Set $X_1 = \bigoplus_{(i,m,\varphi)} E_i[m]$ where the direct sum is over all triples (i, m, φ) such that $i \in I$, $m \in \mathbf{Z}$ and $\varphi : E_i[m] \to X$. Then X_1 comes equipped with a canonical morphism $X_1 \to X$. Given $X_n \to X$ we set $Y_n = \bigoplus_{(i,m,\varphi)} E_i[m]$ where the direct sum is over all triples (i, m, φ) such that $i \in I$, $m \in \mathbf{Z}$, and $\varphi : E_i[m] \to X_n$ is a morphism such that $E_i[m] \to X_n \to X$ is zero. Choose a distinguished triangle $Y_n \to X_n \to X_{n+1} \to Y_n[1]$ and let $X_{n+1} \to X$ be any morphism such that $X_n \to X_{n+1} \to X$ is the given one; such a morphism exists by our choice of Y_n. We obtain a morphism $\mathrm{hocolim} X_n \to X$ by the construction of our maps $X_n \to X$. Choose a distinguished triangle

$$C \to \mathrm{hocolim} X_n \to X \to C[1]$$

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Let $E_i[m] \to C$ be a morphism. Since E_i is compact, Lemma 094A implies the composition $E_i[m] \to C \to \text{hocolim} X_n$ factors through X_n for some n, say by $E_i[m] \to X_n$. Then the construction of Y_n shows that the composition $E_i[m] \to X_n \to X_{n+1}$ is zero. In other words, the composition $E_i[m] \to C \to \text{hocolim} X_n$ is zero. This means that our morphism $E_i[m] \to C$ comes from a morphism $E_i[m] \to X[-1]$. The construction of X_1 then shows that such morphism lifts to $\text{hocolim} X_n$ and we conclude that our morphism $E_i[m] \to C$ is zero. The assumption that $\bigoplus E_i$ generates $\mathcal{D}$ implies that C is zero and the proof is done. $\square$